

Quality Disclosure Programs and Internal Organizational Practices: Evidence from Airline Flight Delays[†]

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Disclosure programs exist in many industries in which consumers are poorly informed about product quality. We study a disclosure program for airline on-time performance, which ranks airlines based on the fraction of their flights that arrive less than 15 minutes late. The program creates incentives for airlines to focus their efforts on flights close to this threshold. We find that firms in this industry are heterogeneous in how they respond to these incentives. Moreover, this heterogeneity correlates with internal firm characteristics. Our findings highlight the importance of interactions between incentives created by a disclosure program and firms' internal organizational practices. (JEL D22, L15, L25, L93)

Quality disclosure programs exist in many industries in which consumers are imperfectly informed about product quality.¹ A growing empirical literature on disclosure programs has documented that disclosure can facilitate better matches between consumers and suppliers and improve the level of quality supplied by firms. However, there is also evidence that disclosure programs can have unanticipated and, in some cases, undesirable outcomes if firms attempt to improve their reported quality through actions that are harmful to consumer welfare. Examples of this type of behavior—which is often termed “gaming”—have been found in a number of contexts including educational accountability and health care.² While there has been a considerable amount of work establishing how disclosure programs may affect the behavior of firms, this work has not explicitly considered how the external

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¹See Dranove and Jin (2010) for a review of the literature on disclosure programs.

²See, for example, Jacob and Levitt (2003); Figlio and Getzler (2006); Cullen and Reback (2006); and Neal and Schanzenbach (2010) on education, and Dranove, Kessler, McClellan, and Satterthwaite (2003); Snyder (2010); and Lu (2012) on health care. See also Kerr (1975).

incentives provided by the program may interact with internal firm characteristics to affect individual firm responses.³

In this paper, we investigate how internal differences between firms influence the responses of these firms to the incentives created by a common government-mandated disclosure program. Like many of the disclosure programs that exist, the program we study reports measures that are based on quality passing a discrete threshold. This structure creates an incentive for firms to focus on improving quality right around the relevant threshold as improvements elsewhere in the quality distribution may not affect the reported metric. While data limitations prevent us from investigating how firms responded to the introduction of this disclosure program, we devise a strategy for estimating whether firms subject to the program respond to the program's incentive to focus on quality improvements close to the threshold. We find—perhaps surprisingly—that some firms do not respond while others appear to respond strongly. We then explore whether firms' responses are correlated with two particular organizational practices and find that they are.

Our empirical setting is the disclosure of airline on-time performance, which has been mandated by the US Department of Transportation (DOT) since 1987. Under this program, the DOT collects information on the arrival delays incurred on every flight operated by every large domestic airline. The DOT then publishes monthly rankings of these airlines based on the fraction of their flights that arrive less than 15 minutes late. The DOT's rankings are frequently cited in the media and are sometimes used by the airlines in their advertising campaigns.⁴ Furthermore, Forbes (2008) presents evidence that airline demand responds to on-time performance.

While the objective of the DOT program was likely to encourage airlines to improve on-time performance on all flights and especially flights with long delays, the design of the program clearly creates an incentive for airlines to reduce delays on specifically those flights that would otherwise arrive just over 15 minutes late—what we term “threshold flights.” These are the flights that—with only a small reduction in delay—would switch from being counted late to being counted on-time. We develop an empirical approach for identifying threshold flights and estimate whether these flights are systematically more likely than any of the airlines' other flights to arrive one or two minutes earlier than expected.⁵ Because we observe tens of thousands of flights each year, we can identify threshold flights very precisely as well as carefully control for unobservables that could lead to reductions in delays on these flights. Specifically, by including airline-airport-day fixed effects in our models, we identify threshold behavior by comparing flights by a given airline arriving at a given airport

³An exception is Jin and Leslie (2009), which shows that franchised and company-owned chain restaurants respond differently to the introduction of hygiene report cards.

⁴More recently, websites that sell airline tickets have begun to publish “on-time” statistics for each flight. These statistics are usually based on the same 15-minute cutoff, but they reflect how often *this flight* has been “on time,” rather than presenting an aggregate statistic for the entire airline. Most of our analysis focuses on a time period before this information was commonly displayed for airline customers at the time of purchase.

⁵We are unfortunately not able to study the impact of the original introduction of the DOT program (and whether it had heterogeneous effects across airlines) because data on on-time performance only became available with the introduction of the program.

on a given day that are predicted to be about 15 minutes late with flights by the same airline arriving at the same airport on the same day that are predicted to be more or less than 15 minutes late.

We combine this empirical approach with information on two airline-level characteristics. The first is the technology used for reporting on-time performance—in this case, whether arrival times are recorded automatically or self-reported by the airline. During the time period we study, airlines varied in how they recorded a flight's on-time performance. Planes that were equipped with a communication technology called Aircraft Communications Addressing and Reporting System (ACARS) directly recorded data about a flight, including its arrival time. Planes that did not have ACARS had their information recorded manually by the airline. We observe both cross- and within-airline variation in reporting technology and investigate whether there is a relationship between reporting technology and threshold behavior.

The second characteristic we consider is the degree to which airline employees are incentivized to care about the firm's performance in the DOT program. Given that threshold flights cannot be identified far in advance, the decision to reduce delays on these flights must be made in real-time and therefore the incentives of employees may be important in this setting. During our sample period, we observe five airlines that introduce employee bonus programs based explicitly on the airline's performance in the DOT ranking.⁶ Under these programs, employees would receive a bonus of between \$65 and \$100 in any month in which the airline as a whole placed at or near the top of the DOT ranking. We investigate whether there is a relationship between the introduction of this type of program and threshold behavior.

Our empirical analysis delivers several findings. First, despite all facing the same incentives to reduce delays below the 15 minute threshold, there is considerable heterogeneity across airlines in whether they engage in threshold behavior. This finding is interesting given that many of the airlines in our data are very similar to each other: they produce similar products, utilize similar aircraft, operate similar types of hub-and-spoke networks, all have unionized employees, and compete with each other for the same customers. The fact that we find heterogeneity in the responses of similar firms to the same program suggests that effects of a disclosure program on firm behavior depend on more than just the incentives created by the program.

Second, we find that both of the firm-specific characteristics that we consider are correlated with threshold behavior. With respect to reporting technology, we find that all of the airlines that report their delays manually engage in threshold behavior, though the magnitude of the effects vary even within the set of manual carriers. We estimate that these airlines' flights that are expected to arrive between 15 and 16 minutes late are between 20 percent and 120 percent more likely than

⁶Knez and Simester (2001) study the effect of one of the bonus programs in our sample (Continental's). They find that overall departure delays decreased after the introduction of the bonus program but they do not investigate threshold behavior. They also cannot address heterogeneity across firms since they only have data from a single airline.

an average flight to arrive exactly one minute earlier than expected. We observe no similar pattern for flights by these airlines that are expected to arrive between 12 and 13 minutes late, between 18 and 19 minutes late, or more than 25 minutes late—all of which could be considered “control groups” for flights in the 15 to 16 minute range. Furthermore, when these airlines switch from manual to automatic reporting, we no longer find evidence that their flights close to the 15 minute threshold are more likely than others to arrive earlier than expected. This last result, in particular, suggests that these airlines were likely misreporting the arrival times of their threshold flights rather than actually reducing the delays on these flights.

With respect to the employee bonus programs, we find strong evidence of threshold behavior for two of the five airlines that introduced such schemes (Continental and Trans World Airlines (TWA)) and weak evidence of threshold behavior for a third airline (American). The remaining two airlines that implemented this type of program show no signs of threshold behavior either before or after the introduction of their program (US Airways and United). Interestingly, Continental and TWA had both manual and automatic planes in their fleets when they introduced their programs while the other three airlines reported fully automatically. Our analysis of Continental and TWA finds that they engage in threshold behavior on both types of planes. This suggests that, at least in these two cases, the bonus programs resulted in actual reductions in delays on threshold flights. More generally, our findings with respect to the bonus programs indicate that—despite the potential for free-riding— incentives based on firm-wide performance can sometimes influence employee behavior. Unfortunately, we do not have detailed enough information about the programs and firms to determine why they affect behavior in some cases but not others. However, we speculate that differences in the structure of the programs (for example, how hard it is to achieve the bonus target) and/or differences in complementary organizational practices adopted by the firm may affect what impact, if any, the bonus programs have on employee incentives.

This paper contributes to the growing literature on disclosure programs. It is the first, we believe, to explicitly focus on documenting firm-specific responses to a common program and relate those responses to internal firm characteristics. Our results indicate that highly similar firms may respond heterogeneously to the incentives created by a given program. Since the objective of most disclosure programs is to incentivize quality improvements by firms while minimizing gaming, understanding the firm-specific characteristics that affect the extent to which a firm will respond to a program (in both desirable and undesirable ways) is clearly critical. More generally, we believe this paper reveals that there could be value gained from integrating insights from information economics and organizational economics. Disclosure programs create market-based incentives for firms to improve quality. Yet, quality production is often the result of large and complex sets of decisions made by individuals within a firm. Thus, understanding how disclosure programs will influence firm behavior requires a nuanced understanding of how firms organize and operate.

The rest of the paper is organized as follows. Section I provides institutional background. Section II describes our data and sample. We outline our empirical approach in Section III. Our results are presented in Section IV. Section V concludes.

I. Institutional Background

A. Reporting of Airline On-Time Performance

Since September 1987, all airlines that account for at least 1 percent of domestic US passenger revenues have been required to submit information about the on-time performance of their flights to the DOT. The reporting requirements have increased over time. Originally, airlines were only required to submit information on their scheduled and actual departure and arrival times and on flight cancellations and diversions.⁷ A January 1995 amendment expanded the requirements to include flights that were delayed or cancelled because of mechanical problems. The same amendment also required that additional data be reported including taxi and airborne times, as well as the aircraft's tail number. Subsequent amendments to the reporting rule required airlines to report delay causes (November 2002) and to report tarmac delays for flights that are cancelled, diverted, or returned to their gate (October 2008).

The DOT uses the data it collects to issue monthly reports that rank airlines based on the percentage of their flights that arrive at the gate with less than 15 minutes of delay. National and local media outlets often report these rankings. A typical news story will mention the percentage of on-time flights for all airlines and may point out which airlines have improved or deteriorated relative to the others, often highlighting which carriers are consistently near the top or the bottom of the ranking. Local media outlets tend to focus on carriers that have a big market share in the local city. It is not uncommon for the media reports to simply refer to flights being "on-time," without explaining the DOT's definition of on-time.

Through our research, we have tried to learn why the DOT chose the fraction of flights less than 15 minutes late to be its reported measure of airline on-time performance, but we have been unable to find an explanation. We can only speculate that the DOT wanted a measure that was easy to compute and understand, that would not be affected by flights with very long delays, and that classified flights with very short delays as "on-time" given that the precise arrival time is merely an estimate. The fraction of flights less than 15 minutes late achieves these objectives; however, it has the drawback of creating a discontinuity in the relationship between actual and reported on-time performance at the 15 minute threshold. Small reductions in delays close to the threshold improve an airline's reported performance while similar reductions in delays that are above or below the threshold do not. While it is possible that this reflects a discontinuity in consumers' marginal disutility of delays, we think it is unlikely. Rather, we suspect that the 15 minute metric was chosen for the types of reasons described above and, while improvements in on-time performance that result from reductions in delays from 15 to 14 minutes are consistent with the incentives created by the DOT's program, they are likely not the type of quality improvement that was envisioned.

⁷The legislation only requires flights to and from 29 of the most congested airports to be included, but all airlines voluntarily report the on-time performance of all their flights—likely because their performance is better at uncongested airports.

An airline that is trying to improve its on-time performance by reducing delays on threshold flights can do so in two ways. It can either try to reduce the actual arrival delay of a flight that is close to the cut-off or it can misreport the arrival time of a flight so that the flight is recorded as arriving less than 15 minutes late. In the first case, the reduction in delay may be achieved either through a reallocation of scarce resources (such as gates or ground crew) from other flights to threshold flights or through the utilization of slack resources (such as getting ground crew to work harder). In the second case, misreporting of the arrival time will only be possible if the airline manually records its on-time performance. It is important to note that, regardless of which mechanism the airline uses, threshold flights cannot be identified in advance of the flight's departure. While an airline can typically predict which flights are likely to be more delayed, on average, than others, it is not able to predict precisely which of its flights on any given day will be 16 versus 14 minutes late. Thus, to the extent that an airline reallocates resources, exerts extra effort or misreports its flight's arrival times, the decision to do so must be made in real-time as the airline obtains information about the flight's expected arrival delay.⁸

B. Reporting Technology

Table 1 shows each carrier's reporting method in March 1998—the earliest month for which this information is available—as well as the dates of any subsequent changes in reporting methods. In 1998, three carriers (Alaska, America West, and Southwest) were reporting their delays manually, four carriers (American, Northwest, United, and US Airways) were reporting automatically, and the remaining three (Continental, Delta, and TWA) were using a combination of manual and automatic reporting. Automatic reporting was facilitated by an aircraft-to-ground communication technology called ACARS. Our understanding is that the primary reasons airlines adopted ACARS were to facilitate real-time aircraft-to-ground communication and to automate the collection of information such as precise departure and arrival times, which is used for crew scheduling and compensation. Given this, it is unlikely that airlines strategically delayed the adoption of ACARS to facilitate misreporting of arrival times. For airlines that had ACARS on some but not all of their planes, we have developed a way to distinguish their manual and automatic aircraft, and we describe this in detail in online Appendix A. Note that most of the manual and combination reporters adopt fully automatic reporting in 2002 or 2003, with the exception of Southwest which does not switch until 2007.

C. Airline Bonus Programs

In February 1995, Continental Airlines implemented the first firm-wide employee bonus program based on the DOT's ranking. Under the program, Continental would pay \$65 to each full-time employee in every month that the airline was among the top five in the DOT's on-time performance ranking. In 1996, the program rules were

⁸Of course, the way in which the airline improves the on-time performance of its threshold flights may have an impact on consumer welfare. We discuss this in Section IV.

TABLE 1—VARIATION IN REPORTING TECHNOLOGY BY CARRIER AND OVER TIME

Carrier	Reporting technology in March 1998	Date of switch:	
		Manual → combination	Combination → automatic
Alaska	Manual	Feb. 2002	—
America West	Manual	Feb. 2002	Jan. 2003
American	Automatic	N/A	N/A
Continental	Combo	N/A	Feb. 2002
Delta	Combo	N/A	Nov. 2003
Northwest	Automatic	N/A	N/A
Southwest	Manual	—	July 2007
TWA	Combo	N/A	—
United	Automatic	N/A	N/A
US Airways	Automatic	N/A	N/A

Notes: Based on information provided in the Air Travel Consumer Reports, which are issued monthly by the Department of Transportation. Note that Southwest Airlines switches from being a fully manual to being a fully automatic reporter in July 2007.

changed to pay each employee \$65 in every month that the airline ranked second or third and to pay \$100 in months that the airline ranked first.⁹ The bonus program was part of a larger turnaround effort called the “Go Forward Plan,” which sought to address poor performance and profitability at the airline (see Bethune and Huler 1999).¹⁰ The two other parts of the “Go Forward Plan” which were also related to improving on-time performance were changes in the flight schedule that increased aircraft turnaround time (i.e., the time between flights) and the replacement or rotation of the senior manager at every airport. While overall improvement in on-time performance after the introduction of the bonus program may be the result of a combination of all three changes, the other components are unlikely to influence threshold behavior. However, increased emphasis within the organization on meeting the DOT’s on-time target may have amplified the effect of the financial incentives that were provided to the employees by communicating that the airline’s management cared about on-time performance.

In June 1996, TWA implemented an employee bonus program that closely resembled Continental’s. Three other airlines introduced similarly structured bonus programs in subsequent years. These were American Airlines in April 2003, US Airways in May 2005, and United Airlines in January 2009. We know that, in all of these cases, the program involved a monthly bonus to all employees in months in which the airline achieved a particular position in the DOT ranking. Unfortunately, we were unable to obtain more systematic and complete information on the specifics of each of the programs. However, based on our reading of the trade press and annual reports, we have learned that the three later programs only rewarded employees if the airline achieved first or second place in the rankings.¹¹ We also know that American Airlines introduced its bonus program in conjunction with a large negotiated wage cut during the industry’s downturn and that US Airways introduced

⁹Our information on these changes comes from newspaper reports. These reports do not include the precise date on which these changes were implemented.

¹⁰In 1994, Continental had the worst average on-time performance ranking among the ten reporting airlines.

¹¹US Airways only rewarded first place, while the other two rewarded first and second place.

its program around the same time that it merged with America West. Finally, while American Airlines had a history of good on-time performance when its program was introduced—having ranked third, on average, during the year prior to the introduction—the other four airlines introduced their programs at times where they had a poor record in on-time performance.

It is worth pointing out that these bonus programs represent a substantial financial commitment by the airlines. For example, when Continental introduced its program in February 1995, it had over 31,000 full-time employees. Thus, a \$65 bonus to all full-time employees would amount to a total payment of a little over \$2 million. United had 43,000 full-time employees when it introduced its bonus program in January 2009. A \$100 bonus for each full-time employee would amount to \$4.3 million. In both cases, this would translate to about \$0.85 per passenger in the respective months. The fact that airlines were willing to pay such amounts to reward improvements in on-time performance indicates that delays are quite costly for airlines affecting both willingness to pay and operating costs.

II. Data

A. Data and Sample

Our empirical analysis uses the flight-level data on on-time performance collected by the DOT. Our regression sample includes domestic flights operated by the ten airlines that met the reporting requirement in 1995: Alaska Airlines, America West, American Airlines, Continental Airlines, Delta Air Lines, Northwest Airlines, TWA, Southwest Airlines, United Airlines, and US Airways.¹² We estimate our main empirical specifications for the period 1995–2000. In 1995, the DOT began collecting data on taxi times, which we require for our empirical analysis. We choose not to include post-2000 data in our main specifications for two reasons. First, after September 2001, both the supply-side and the demand-side incentives for on-time performance likely changed substantially (see, for example, Berry and Jia 2010). Second, the volume of the available data is so large that we are unable to estimate regressions that include all the airlines for a longer time period. However, we use the later data when we explore the bonus programs that are introduced later in the sample as well as when we explore airlines' transitions from manual to automatic reporting.

Due to the large volume of data, we take a random sample of flights by restricting to every ninth day of the year. In addition, we drop flights that meet any of the following conditions: depart more than 15 minutes early (since we suspect this may represent a rescheduled flight), arrive more than 90 minutes early, depart on what appears to be the following calendar day, have a taxi-out or taxi-in time of more than 60 minutes, have missing values for their scheduled arrival or departure times, have a distance of less than 25 miles, or operate fewer than 20 times during the quarter. Our final sample for the 1995 to 2000 period includes over 3.3 million flights. Table 2 presents summary statistics for the main regression sample. The average

¹²TWA was acquired by American Airlines in 2001. America West merged with US Airways in 2005.

TABLE 2—SUMMARY STATISTICS FOR MAIN REGRESSION SAMPLE

	Mean	SD	Min	Max
<i>1995–2000 sample</i> (3,326,681 observations)				
Arrival delay (min)	7.50	28.23	–86	1,251
Dummy for arrival delay of 15 minutes or more	0.21	0.41	0	1
Taxi-in time (min)	5.78	3.93	1	60
Departure delay (min)	8.79	25.91	–15	1,246
Taxi-out time (min)	14.36	7.58	1	60
Air time (min)	101.76	64.71	20	641

Notes: Includes flights by Alaskan, America West, American, Continental, Delta, Northwest, TWA, United, and US Airways. TWA is acquired by American in 2001. America West merges with US Airways in 2005.

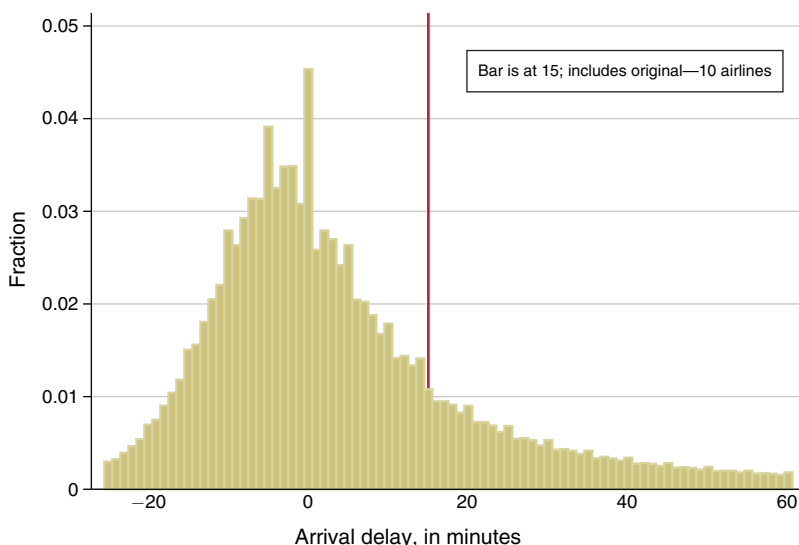


FIGURE 1A. DISTRIBUTION OF ARRIVAL DELAYS, 1995–2000.
TEN US CARRIERS THAT MET ORIGINAL DOT REPORTING REQUIREMENT

arrival delay in the sample is 7.5 minutes. Twenty-one percent of all flights arrive 15 or more minutes late and thus are considered late under the DOT's definition.

B. Histograms of Arrival Delays

Figures 1A and 1B show the distribution of arrival delays for the ten carriers in our sample for the periods 1995–2000 and 2002–2010, respectively. The histogram for the earlier period is fairly smooth, peaks at 0 and has small spikes that occur at five minute intervals (e.g., at –5, 0, 5, 10, etc. ...); however, instead of there being a spike at 15 minutes, there is a spike at 14 minutes. In contrast, the histogram for the later period is completely smooth. In Figures 2A–2D, we show the distribution of arrival delays by reporting method. Figure 2A shows the distribution of arrival delays for the four fully automatic carriers. Their histogram is smooth with a peak around –5 and no apparent spike at 14 minutes. Figure 2B shows arrival delays for the three airlines that used a combination of manual and automatic reporting during

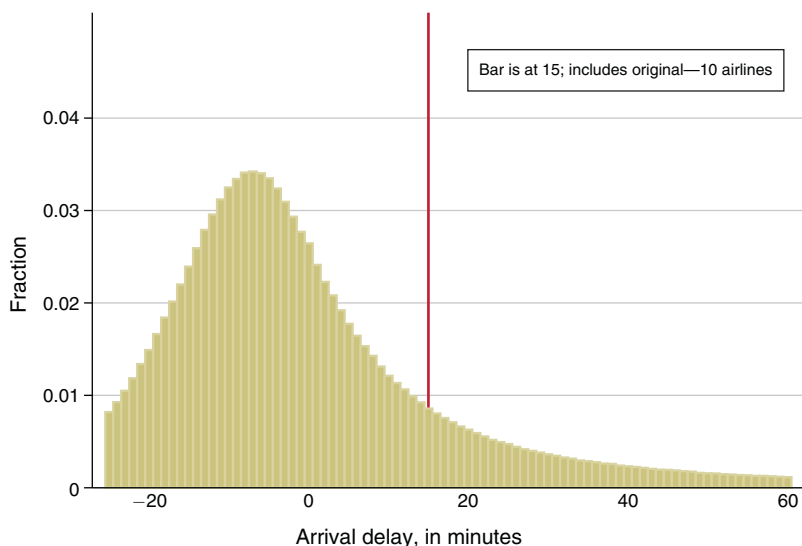


FIGURE 1B. DISTRIBUTION OF ARRIVAL DELAYS, 2002–2010.
TEN US CARRIERS THAT MET ORIGINAL DOT REPORTING REQUIREMENT

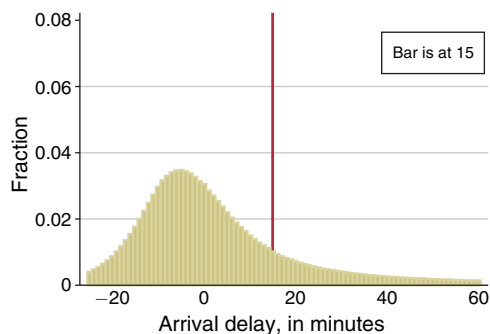


FIGURE 2A. AUTOMATIC REPORTERS

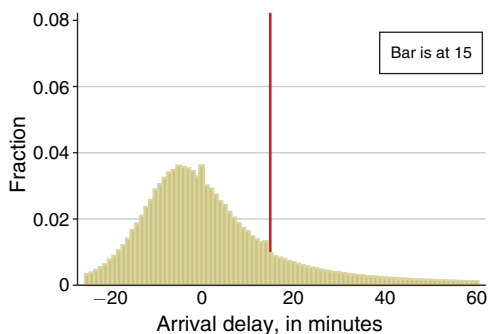


FIGURE 2B. COMBINATION REPORTERS

Note: Automatic reporters include America, Northwest, United, and US Airways.

Note: Combination reporters include Continental, Delta, and TWA.

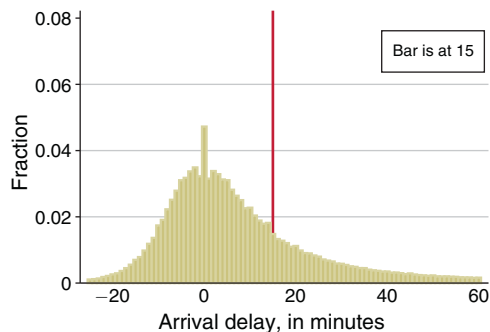


FIGURE 2C. MANUAL REPORTERS,
EXCLUDING SOUTHWEST

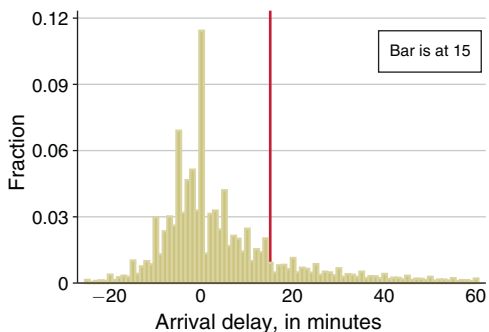


FIGURE 2D. SOUTHWEST AIRLINES

Note: Manual reporters other than Southwest include Alaska and America West.

Note: Note that the scale on Figure 2D is slightly different than the other three.

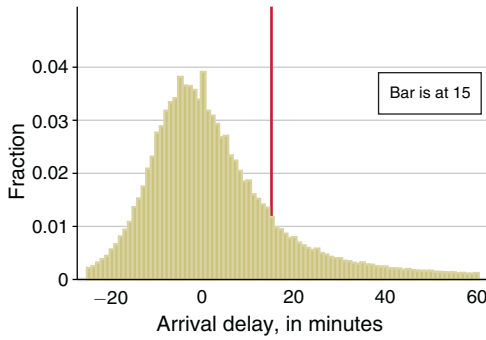


FIGURE 3A. CONTINENTAL AIRLINES, BEFORE

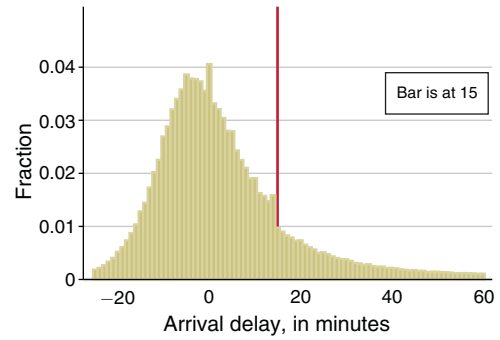


FIGURE 3B. CONTINENTAL AIRLINES, AFTER

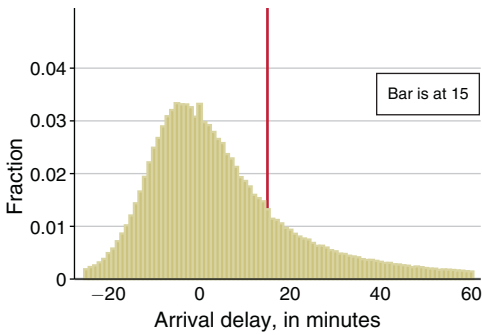


FIGURE 4A. TWA, BEFORE

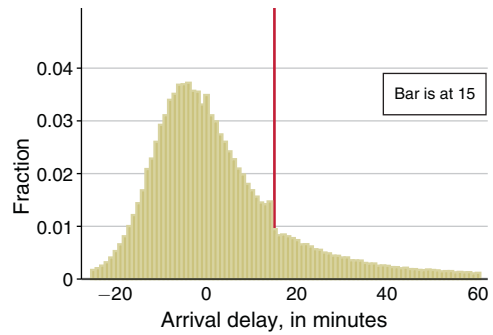


FIGURE 4B. TWA, AFTER

FIGURES 3–7. DISTRIBUTION OF ARRIVAL DELAYS 18 MONTHS
BEFORE AND AFTER INTRODUCTION OF BONUS PROGRAMS

(Continued)

this period. Their histogram is also quite smooth; however, it shows distinct spikes at 0 and at 14 minutes. Figure 2C shows the distribution of arrival delays for two of the three carriers which reported their on-time data manually (Alaska Airlines and American West). This histogram is also reasonably smooth, but has a large spike at zero and a much smaller, but noticeable spike at 14 minutes. Finally, in Figure 2D, we show arrival delays for Southwest Airlines. Southwest's histogram has a large spike at zero (with almost 12 percent of flights arriving with exactly zero minutes delay) and spikes at all of the five minute intervals except 15, and a spike at 14 minutes. This suggests that Southwest Airlines was rounding flight delays to the nearest multiple of five, except at the 15 minute threshold where the rounding appears to have occurred at 14.

In Figures 3–7, we compare the before-and-after distributions of arrival delays for each of the airlines that introduced employee bonus programs. Figures 3A and 3B show arrival delays for Continental Airlines and reveal that, after the introduction of the bonus program, there is an increase in the number of flights that arrive exactly 14 minutes late and a decrease in the number of flights that arrive 15 or 16 minutes late. Figures 4A and 4B plot analogous histograms for TWA and show a very similar pattern. Figures 5A and 5B plot the arrival delay distribution for American Airlines and show a very small discontinuity around the 14 minute mark. The analogous figures for US Airways and United Airlines show no apparent discontinuity at 14 minutes.

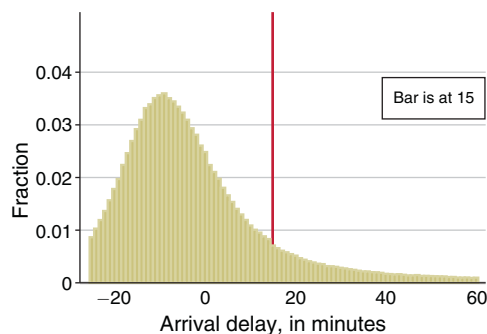


FIGURE 5A. AMERICAN AIRLINES, BEFORE

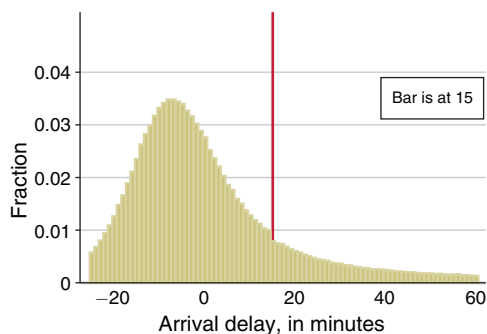


FIGURE 5B. AMERICAN AIRLINES, AFTER

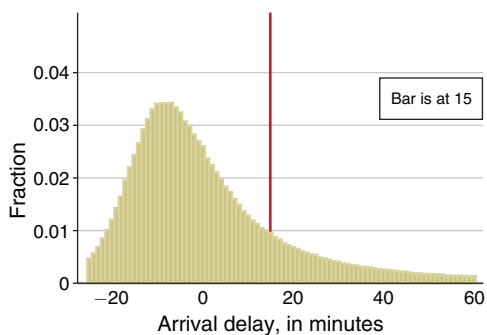


FIGURE 6A. US AIRWAYS, BEFORE

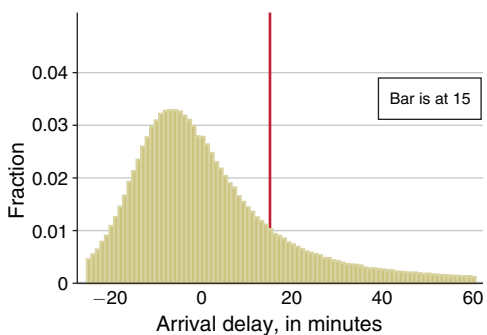


FIGURE 6B. US AIRWAYS, AFTER

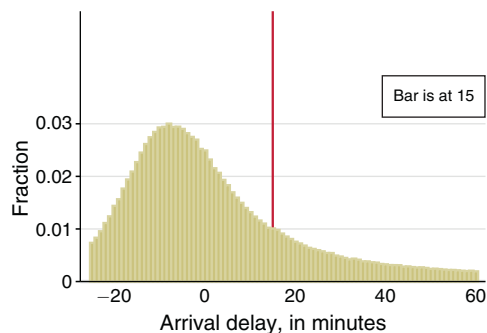


FIGURE 7A. UNITED AIRLINES, BEFORE

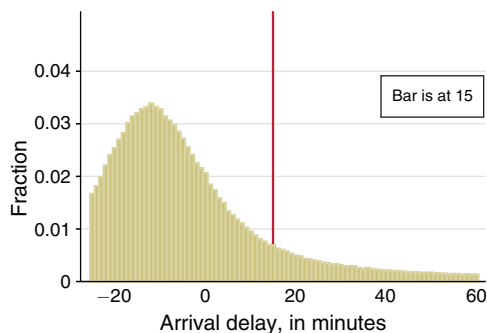


FIGURE 7B. UNITED AIRLINES, AFTER

FIGURES 3–7. DISTRIBUTION OF ARRIVAL DELAYS 18 MONTHS
BEFORE AND AFTER INTRODUCTION OF BONUS PROGRAMS (*Continued*)

III. Empirical Approach

A. Overview of the Empirical Approach

Our objective is to determine whether airlines engage in threshold behavior. To do this, we have to estimate whether airlines systematically reduce delays on flights that they expect to arrive with a delay of just over 15 minutes. This requires two things. First, we need to identify flights that an airline expects to arrive close to the 15 minute threshold. Second, we need to measure whether these flights arrive

with a delay that is less than what it otherwise would have been. This requires a counterfactual measure of the flight's delay absent the incentive arising from the threshold effect.

Because our data allow us to observe the various stages of a flight—departure from the gate, take-off, landing, and arrival at the gate—we can construct a flight's expected delay at each stage and, at any given stage, we can identify those flights whose expected delay is close to 15 minutes. We can then investigate whether—in *subsequent* stages of the flight—airlines reduce delays on specifically those flights that are expected to be around 15 minutes late. Furthermore, we have several ways of determining the counterfactual delay that these flights would have had in the subsequent stages. First, we can look at flights just outside the critical threshold. We can assume that—absent the incentives created by the program's design and controlling for observable flight characteristics—an airline's behavior with respect to flights that are expected to arrive 15 minutes late should be similar to its behavior with respect to flights that are expected to arrive, say, 12 or 18 minutes late. Second, we can compare flights in the 15 minute range to flights with very long expected delays. If the costs of delays are convex, then airlines should have the greatest incentives to reduce delays on those flights. If we find that airlines are more likely to reduce delays on flights that they expect to arrive close to the 15 minute threshold than on flights that they expect to arrive with 12 or 18 minutes of delay, or with very long delays, we will interpret this as evidence that these airlines are engaging in threshold behavior. It is also worth pointing out that airlines cannot engage in *ex ante* behavior that aims to reduce delays specifically on those flights that they expect to arrive right around 15 minutes late since they simply do not know in advance which flights these will be.

B. Regression Analysis

While delays can occur at any stage of a flight, we expect that airlines attempting to speed up threshold flights will be more likely to do so during the later stages. This is because, as the flight progresses, the airline knows the delay incurred so far and can therefore more precisely predict the total delay the flight will have. For any given predicted level of delay, reducing the amount of noise associated with that prediction increases the likelihood that the airline's effort to reduce the delay of a flight to just below 15 minutes will be successful. Based on this, we focus on estimating systematic reductions in delays during the final phase of the flight—i.e., when it is taxiing in to its arrival gate. Focusing on this last stage of the flight also has the advantage that we can predict taxi times with much less noise than airborne time because airborne time depends on wind patterns that change from day to day and on which we do not have any data.

In order to identify threshold flights, we construct each flight's expected delay at the time that its wheels touchdown at the arrival airport. We take the flight's observed wheels-down time and add to it the median taxi-in time for that flight in the quarter.¹³ This gives us a predicted arrival time for the flight. The difference

¹³ We identify a flight as a unique combination of airline, flight number, departure airport, and arrival airport.

between the predicted arrival time and the scheduled arrival time is the flight's predicted delay.¹⁴ We then construct a series of dummy variables for each level of predicted delay, in one minute increments. For example, we construct a dummy variable that equals 1 if a flight's predicted delay is greater than or equal to 10 minutes and less than 11 minutes. We construct another dummy variable that is equal to 1 if a flight's predicted delay is greater than or equal to 11 minutes and less than 12 minutes. Flights with predicted delays of greater than 25 minutes are grouped together in the top category while flights with predicted delays of less than 10 minutes are used as the excluded category. Thus, we define 16 different predicted delay "bins." In order to measure whether flights that are expected to arrive 15–16 minutes late are arriving just under 15 minutes late, we construct a dummy variable that equals one if a flight arrives exactly one minute earlier than predicted. This is our main dependent variable. We regress this dummy variable on the predicted delay bins described above and a set of control variables which include carrier-arrival airport-day fixed effects, a dummy for the departure airport being a hub, controls for two distance categories (500–1,500 miles and greater than 1,500 miles), and dummies for each (actual) arrival hour. We cluster our standard errors at the arrival airport-day level.

The coefficients on the delay bins represent the change in the probability that a flight in a given predicted delay bin will arrive exactly one minute earlier than predicted, relative to the probability of arriving one minute early for flights with a predicted delay of less than 10 minutes (the excluded category). Our primary interest is in testing whether flights with predicted delay right around the 15 minute threshold are systematically more likely to arrive exactly one minute earlier than predicted, compared to flights that are a bit further above or below the 15 minute threshold. Because we include carrier-airport-day fixed effects, our coefficients are identified by variation in predicted delays across an airline's flights that arrive at a given airport on a given day. This variation results from differences in the delays that flights incur *prior* to arrival which will be driven by factors at the flights' respective departure airports and in the air.

The key identifying assumption of the model is that there are no observable factors that are correlated with a flight having a predicted delay just above the 15 minute threshold and that would also result in the flight being more likely than flights just outside this range to arrive one minute earlier than predicted. To present further evidence that our findings indeed represent a response to the structure of the DOT program, we also estimate the same regressions using the probability that a flight arrives exactly two minutes early as the dependent variable. In these regressions, we test whether flights with a predicted delay of 16–17 minutes are systematically more likely than other flights to arrive two minutes earlier than predicted. All of the results using the two-minute early variable are presented in the online Appendix.

¹⁴ For example, consider a flight by Delta Air Lines between Boston and Atlanta in March of 1997. Suppose that it has a scheduled arrival time of 4:30 PM. If its wheels-down time is 4:36 PM and Delta's median taxi-in time for this flight in this quarter is four minutes, then the flight's predicted arrival time is 4:40 PM and its predicted delay is 10 minutes.

To investigate the existence and sources of heterogeneity in airlines' threshold behavior, we construct different sets of predicted delay bins for different groups of flights and/or airlines. For example, to estimate the threshold behavior of each carrier, we include a separate set of predicted delay bins for each of the ten airlines in the sample. To investigate whether an airline's behavior is different on its manual and automatic planes, we construct a separate set of predicted delay bins for each combination of reporter's manual flights and automatic flights. To investigate whether an airline's behavior is affected by the introduction of an employee bonus program, we construct a separate set of predicted delay bins for the airline's flights before and after the introduction of its program. In general, our empirical approach provides a flexible way of investigating the extent to which threshold behavior varies across different airlines, flights, or time periods.

IV. Results

Our regression results are presented in Tables 3 through 7. Most tables show the results from a single regression with the columns showing the coefficients for the different sets of predicted delay bins included in the regression. We begin by presenting a baseline regression in which we combine all airlines in a single set of predicted delay bins (Table 3). We then explore whether airlines are heterogeneous in how they behave with respect to their threshold flights by including separate predicted delay bins for each airline (Table 4). After finding that they are and that the heterogeneity appears to relate to reporting technology and employee incentives, we then investigate the role of these two characteristics in more detail (Tables 5 through 7). In online Appendix B, we present the results of estimating each model using, as the dependent variable, a dummy for arriving two minutes earlier than predicted. In all cases, the results of the two-minute analysis are consistent with those presented here.

A. Overall Effects

Table 3 shows the results of estimating the regression described in Section III with a single set of predicted delay bins. The coefficient on the 15–16 minute bin is estimated to be 0.074 which indicates that flights that are predicted to arrive between 15 and 16 minutes late are 7.4 percentage points more likely than flights in the excluded category to arrive exactly one minute earlier than predicted. The coefficient is statistically and economically significant.¹⁵ Averaging across all flights in the sample, the probability of arriving one minute earlier than predicted is 0.21. Thus, our estimate of 0.074 represents about a 35 percent increase in the likelihood of being one minute

¹⁵ Because airlines may have more precise estimates of a flight's taxi-in time than the estimates we use to construct our predicted delay measures, we expect that we will misclassify some flights. For example, our predicted delay calculation could classify a flight that the airline expects to arrive 20 minutes late in the 19 minute predicted delay bin. If we misclassify some flights that an airline expects to arrive 15 minutes late, then we could find a positive coefficient on the predicted delay bins on either side of the 15–16 minute bin that we focus on, as we do in Table 3. However, in Table 3 and all subsequent tables, these coefficients are much smaller in magnitude than the estimates on the 15 minute bin.

TABLE 3—PROBABILITY OF ARRIVING ONE MINUTE EARLIER THAN PREDICTED, 1995–2000, ALL REPORTING CARRIERS

Dependent variable	= 1 if flight arrives one minute earlier than predicted
[10,11) min	−0.004 ⁺ (0.002)
[11,12) min	0.022** (0.002)
[12,13) min	−0.000 (0.002)
[13,14) min	0.003 (0.002)
[14,15) min	0.005 ⁺ (0.002)
[15,16) min	0.074** (0.003)
[16,17) min	0.008* (0.002)
[17,18) min	−0.009** (0.002)
[18,19) min	0.003 (0.003)
[19,20) min	−0.000 (0.003)
[20,21) min	0.002 (0.003)
[21,22) min	0.029** (0.003)
[22,23) min	0.000 (0.003)
[23,24) min	0.012** (0.003)
[24,25) min	0.007 ⁺ (0.003)
≥25 min	0.006** (0.001)
<i>p-values from hypothesis tests of equality of coefficients</i>	
Bin15 = Bin12	0.00
Bin15 = Bin18	0.00
Bin15 = Bin25	0.00
Pr(1 min early)	0.21

Notes: Standard errors are in parentheses and clustered at the level of the arrival airport-day. Coefficients represent the change in the probability of a flight arriving one minute earlier than predicted relative to flights with predicted delay less than 10 minutes. The regression contains 3,326,681 observations.

** Significant at the 1 percent level.

* Significant at the 5 percent level.

⁺ Significant at the 10 percent level.

early. No other level of predicted delay has a coefficient this large. In this and most subsequent specifications, we test whether the coefficient on the 15–16 minute bin is equal to the coefficient on the 12–13 minute bin, the coefficient on the 18–19 minute bin and the coefficient on the 25+ minute bin. The *p*-values presented in Table 3 indicate that we can reject each of these hypotheses.

TABLE 4—PROBABILITY OF ARRIVING ONE MINUTE EARLIER THAN PREDICTED, 1995–2000, CARRIER-SPECIFIC ESTIMATES

Carrier	Coefficient on [15,16] bin	Pr(1 min early)	Number (percent) flights in [15,16] bin	Reporting technology	Bonus program during 1995–2000
American	0.006 (0.007)	0.178	3,224 (0.86 percent)	Automatic	No
Alaska	0.065** (0.012)	0.231	1,462 (1.5 percent)	Manual	No
Continental	0.101** (0.009)	0.213	2,484 (0.95 percent)	Combination	Yes
Delta	0.017** (0.005)	0.205	6,887 (1.2 percent)	Combination	No
America West	0.045** (0.011)	0.213	1,701 (1.3 percent)	Manual	No
Northwest	0.015 (0.008)	0.191	2,738 (0.81 percent)	Automatic	No
TWA	0.073** (0.012)	0.204	1,682 (0.95 percent)	Combination	Yes
United	0.007 (0.006)	0.204	4,503 (0.96 percent)	Automatic	No
US Airways	0.003 (0.006)	0.224	4,676 (1.0 percent)	Automatic	No
Southwest	0.316** (0.007)	0.227	5,678 (1.1 percent)	Manual	No

Notes: Standard errors are in parentheses and clustered at the level of the arrival airport-day. The second column of the table shows the coefficient on the [15,16] bin for each airline. The coefficient represents the change in the probability of arriving one minute earlier than predicted for flights with predicted delay between 15 and 16 minutes, relative to flights with predicted delay less than 10 minutes. The third column shows the fraction of each airline's flights that arrive one minute earlier than predicted. The fourth column shows the number (percent) of each airline's flights that are predicted to arrive between 15 and 16 minutes late. The regression contains 3,326,681 observations.

** Significant at the 1 percent level.

* Significant at the 5 percent level.

+ Significant at the 10 percent level.

B. Carrier-Specific Estimates

We now estimate our main regression with a separate set of predicted delay bins for each airline. In the first column of Table 4, we present the coefficient estimate on the 15–16 minute bin for each airline. We also show the overall fraction of each airline's flights that arrive one minute earlier than predicted (the second column) and the number (and percent) of each airline's flights that are predicted to be between 15 and 16 minutes late (the third column). The final two columns of the table indicate each airline's reporting technology and whether it had an employee bonus program during this period.

The estimates in this table reveal that there is considerable heterogeneity across airlines in the extent to which they manipulate their threshold flights.¹⁶ Four of the airlines in our sample (American, Northwest, United, and US Airways) exhibit no

¹⁶ It is worth noting that while we find that airlines' threshold behavior varies, the ten airlines in our sample are quite similar in terms of the fraction of their flights that arrive one minute early (about 20 percent) and the fraction of their flights that are predicted to be between 15 and 16 minutes late (about 1 percent).

evidence of threshold behavior. All four of these airlines are automatic reporters and do not have a bonus program in place during this period. Six airlines (Alaska, Continental, Delta, America West, TWA, and Southwest) show evidence of threshold behavior. The magnitude of the point estimate varies considerably even among these six, ranging from 0.017 for Delta to 0.316 for Southwest. All of these airlines are manual or combination reporters during this time period and Continental and TWA introduce employee bonus programs during these years.

C. The Role of Reporting Technology

We now explore the role of reporting technology in more detail. We begin by noting that the estimates in Table 4 indicate that all three of the fully manual reporters (Alaska, America West, and Southwest) engage in threshold behavior. In addition, the coefficient estimates on these airlines' other predicted delay bins (not reported but available upon request) reveal that these three carriers also engage in rounding of their delays, typically to multiples of five. We find positive and statistically significant coefficients on the [11,12) bin for Alaska and Southwest and positive and statistically significant coefficients on the [21,22) bin for all three of these airlines. If their pattern was to round to the nearest five-minute interval, we would expect a positive coefficient on the [16,17) bin; yet, consistent with the incentives stemming from the program's design, we instead see the positive coefficient on the [15,16) bin. In all cases, the point estimates on the [11, 12) and [21,22) are smaller in magnitude than the estimates on the [15,16) bin.

To further investigate whether the patterns we estimate are consistent with misreporting of arrival times, we examine what happens when these three manual carriers switch to combination or automatic reporting. Alaska and America West switch from manual to combination reporting in February 2002 and America West switches to fully automatic reporting in January 2003, while Alaska remains a combination reporter. Southwest switches from manual to fully automatic reporting in July 2007. Table 5 presents the results of estimating our main specification for these three airlines for the years 2002 to 2008.¹⁷ The first two columns show that neither Alaska nor America West engages in threshold behavior during the later period. The coefficients on the [15,16) minute bin are statistically not distinguishable from zero. In addition, there is no longer any evidence of rounding by these airlines as the coefficients on the [11,12) and [21,22) bins (not reported) are also indistinguishable from zero. The third and fourth columns show the coefficients for Southwest before and after it switches to automatic reporting. Prior to switching, we continue to see evidence of rounding and threshold behavior by Southwest. After Southwest begins reporting automatically, we no longer see evidence of rounding or threshold effects. Taken together, the findings in Tables 4 and 5 indicate that manual reporting facilitates misreporting of arrival times and that airlines use this as a way to move flights over the threshold for being considered on-time.

¹⁷ Due to space constraints, we only present a subset of the 16 coefficients. The full set of results is available upon request.

TABLE 5—PROBABILITY OF ARRIVING EXACTLY ONE MINUTE EARLIER THAN PREDICTED, 2002–2008, CARRIERS THAT SWITCH REPORTING TECHNOLOGY

Dependent variable	= 1 if flight arrives one minute earlier than predicted			
	Southwest			
	America West	Alaska	Manual period	Automatic period
[12,13) min	0.013 (0.011)	−0.004 (0.009)	−0.060** (0.004)	−0.001 (0.007)
[13,14) min	0.006 (0.011)	−0.016 (0.009)	−0.010* (0.004)	0.001 (0.007)
[14,15) min	0.007 (0.011)	0.003 (0.009)	0.007 (0.004)	0.022* (0.007)
[15,16) min	−0.007 (0.012)	−0.003 (0.010)	0.118** (0.005)	0.006 (0.007)
[16,17) min	0.007 (0.012)	−0.027* (0.009)	0.128** (0.006)	0.024* (0.008)
[17,18) min	0.018 (0.013)	−0.008 (0.010)	−0.064** (0.005)	0.017+ (0.008)
[18,19) min	0.014 (0.013)	0.009 (0.011)	−0.029** (0.005)	0.000 (0.008)
[19,20) min	0.013 (0.013)	−0.015 (0.011)	0.007 (0.006)	−0.000 (0.009)
≥25 min	0.010* (0.004)	0.008* (0.003)	0.003 (0.002)	0.001 (0.002)

Notes: Standard errors are in parentheses and clustered at the level of the arrival airport-day. Columns display select coefficients from a single regression that includes carrier-arrival airport-day fixed effects and arrival hour and hub controls. Coefficients represent the change in the probability of a flight arriving one minute earlier than predicted, relative to flights with a predicted delay of less than 10 minutes. The regression contains 2,904,668 observations.

** Significant at the 1 percent level.

* Significant at the 5 percent level.

+ Significant at the 10 percent level.

D. The Role of Employee Bonus Programs

Between 1995 and 2009, five airlines introduced employee incentive programs based on the airline's performance in the DOT ranking. The estimates in Table 4 for the 1995 to 2000 period show evidence of threshold behavior by Continental and TWA, the two airlines that introduced bonus programs during that period. However, both of these airlines were combination reporters at the time; thus, without differentiating between their automatic and manual planes, we do not know whether Continental and TWA's threshold behavior is affected by their reporting technology, their bonus program, or both.

To disentangle these effects, Table 6 presents results for Continental and TWA during this time period, separating their manual and automatic planes and also separating TWA's flights before and after it introduced its bonus program.¹⁸ We do not observe data on taxi times prior to the introduction of Continental's bonus program which is why we only estimate the effect during the postperiod for this carrier. For

¹⁸ As in the other tables, we only present a subset of the 16 coefficients. The full set of results is available upon request.

TABLE 6—PROBABILITY OF ARRIVING ONE MINUTE EARLIER THAN PREDICTED, 1995–2000, COMBINATION REPORTERS

Dependent variable	= 1 if flight arrives one minute earlier than predicted							
	Continental		TWA prebonus		TWA postbonus		Delta	
	Manual	Automatic	Manual	Automatic	Manual	Automatic	Manual	Automatic
[12,13) min	0.015 ⁺ (0.007)	0.017** (0.004)	0.010 (0.028)	−0.005 (0.009)	0.044 ⁺ (0.019)	0.019** (0.006)	−0.005 (0.003)	0.010** (0.003)
[13,14) min	0.047** (0.008)	0.035** (0.004)	−0.017 (0.029)	−0.002 (0.009)	0.024 (0.018)	0.039** (0.006)	−0.000 (0.004)	0.017** (0.003)
[14,15) min	0.034** (0.008)	0.049** (0.005)	0.037 (0.031)	0.001 (0.010)	0.049 ⁺ (0.020)	0.039** (0.006)	0.007 (0.004)	0.015** (0.003)
[15,16) min	0.152** (0.010)	0.083** (0.005)	0.080 ⁺ (0.036)	0.017 (0.010)	0.221** (0.024)	0.078** (0.007)	0.013** (0.004)	0.021** (0.003)
[16,17) min	−0.017 ⁺ (0.009)	−0.009 ⁺ (0.005)	0.042 (0.032)	0.001 (0.010)	−0.048* (0.018)	−0.025** (0.006)	0.004 (0.004)	0.013** (0.003)
[17,18) min	−0.002 (0.009)	−0.006 (0.005)	−0.016 (0.030)	−0.011 (0.010)	−0.047 ⁺ (0.020)	−0.013 ⁺ (0.006)	−0.003 (0.004)	0.010* (0.003)
[18,19) min	−0.014 (0.010)	0.004 (0.005)	−0.004 (0.036)	−0.004 (0.011)	0.006 (0.024)	−0.001 (0.007)	−0.003 (0.004)	0.019** (0.003)
≥25 min	0.011** (0.003)	0.012** (0.002)	0.004 (0.009)	0.007* (0.003)	0.003 (0.006)	0.009** (0.002)	0.002 (0.001)	0.015** (0.001)
<i>p-values from hypothesis tests of equality of coefficients</i>								
Bin15 = Bin12	0.00	0.00	0.13	0.10	0.00	0.00	0.00	0.00
Bin15 = Bin18	0.00	0.00	0.10	0.14	0.00	0.00	0.01	0.55
Bin15 = Bin25	0.00	0.00	0.04	0.31	0.00	0.00	0.01	0.04
Pr (1 min early)	0.22	0.21	0.21	0.20	0.20	0.20	0.20	0.21

Notes: Standard errors are in parentheses and clustered at the arrival airport-day. Columns show select coefficients from a single regression with carrier-arrival airport-day fixed effects, arrival hour, and hub controls. Coefficients represent the change in the probability of a flight arriving one minute earlier than predicted relative to flights with predicted delay of less than 10 minutes. The regression includes flights by Delta, Continental, and TWA on every other day between 1995 and 2000. The regression contains 4,485,758 observations.

** Significant at the 1 percent level.

* Significant at the 5 percent level.

⁺ Significant at the 10 percent level.

comparison purposes, we also include Delta in the table. Delta is also a combination reporter during this period; however, unlike Continental and TWA, Delta does not have an employee bonus program in place during this time.

Looking first at Continental, we find evidence of threshold behavior for both its manual and its automatic planes. The estimates in the first two columns imply that Continental's manual planes that are predicted to be 15 to 16 minutes late are 15.2 percentage points more likely to arrive one minute earlier than predicted and its automatic planes are 8.3 percentage points more likely. This is compared to the overall probability of arriving one minute early for these groups of about 20 percent. The coefficients on all of the other predicted delay bins for Continental are substantially smaller in magnitude.

With respect to TWA, we find evidence of threshold effects for TWA on its manual planes prior to the introduction of its bonus program and on both types of planes after the introduction. The coefficient on TWA's [15,16) bin for its manual flights is 0.08, which implies that these flights are 8 percentage points more likely than flights in the excluded category to arrive 1 minute earlier than predicted. The coefficient is significantly different from zero only at the 10 percent level, but this is

not surprising given that there are only about 345 TWA flights by manual planes in the prebonus period that fall into this predicted delay bin. In TWA's postperiod, we estimate that its manual planes that are predicted to be between 15 and 16 minutes late are 22.1 percentage points more likely to arrive 1 minute early than its flights in the excluded category. This is a large effect given that TWA's flights, on average, arrive 1 minute earlier than predicted about 20 percent of the time. For automatic planes, the estimated effect is 7.8 percentage points, which is considerably smaller than the estimate for the manual planes, but still a sizeable effect given the underlying probability.

The last two columns of the table show the results for Delta. We estimate statistically significant but very small coefficients on the 15–16 bin for both Delta's manual and automatic planes (1.3 and 2.1 percentage points, respectively). For Delta's manual planes, the [15,16) bin is the only one with a positive coefficient suggesting some rounding or misreporting of threshold flights. For Delta's automatic planes, we find positive and statistically significant coefficients on many of the other predicted delay bins as well. We can reject the hypothesis that the [15,16) bin coefficient is equal to the coefficient on the [12,13) and the coefficient on the greater than 25 minute bin. However, we cannot reject the equality of the [15,16) coefficient and the coefficients on several of the other predicted delay bins. Thus, the evidence of threshold behavior by Delta is weak.

Overall, the results in Table 6 indicate that—for Continental and TWA—the introduction of their employee bonus programs is correlated with an increase in threshold behavior. For manual planes, much of the behavior we observe may be due to misreporting. For automatic planes, however, the results indicate actual reductions in the delays of threshold flights. For TWA, we can observe that the carrier's behavior changed substantially after the bonus program was introduced. While we do not observe taxi times prior to the bonus program for Continental, we do observe arrival delays. Recall that in the histograms of arrival delays, we observed a very similar change for Continental after the introduction of the bonus program as for TWA and no spike at 14 in Continental's histogram prior to the introduction of its program.

We now turn to the bonus programs which were introduced in 2003 (American), 2005 (US Airways), and 2009 (United). Each was introduced by an airline that used fully automatic reporting. To investigate whether these programs had effects similar to the two earlier programs, we estimate a regression for each carrier using data from 12–18 months before and after the introduction of their program. We present these results in Table 7. For American Airlines, our estimates suggest that prior to the introduction of its program, its flights in the [15,16) minute bin were 1.2 percentage points more likely than flights in the omitted category to arrive one minute early. After the introduction, the estimated effect increases to 2.2 percentage points. We can reject the equality of this coefficient and those on the control bins at the 5 percent level and we can reject the equality of this coefficient on the coefficient on the [15,16) minute bin from the prebonus period at the 10 percent level. While this provides some evidence of threshold behavior by American after the introduction of its bonus program, the magnitude of the effect is very small compared to what we estimated for Continental and TWA. The remaining results in the table indicate that

TABLE 7—PROBABILITY OF ARRIVING EXACTLY ONE MINUTE EARLIER THAN PREDICTED, LATER BONUS PROGRAMS

Dependent variable	= 1 if flight arrives one minute earlier than predicted					
	American Airlines		US Airways		United Airlines	
	Prebonus	Postbonus	Prebonus	Postbonus	Prebonus	Postbonus
[12,13) min	−0.001 (0.005)	0.005 (0.003)	−0.001 (0.006)	0.004 (0.003)	−0.005 (0.006)	0.011 (0.006)
[13,14) min	0.008 (0.005)	0.009* (0.003)	0.002 (0.006)	0.003 (0.003)	−0.013+ (0.006)	0.002 (0.007)
[14,15) min	0.014* (0.005)	0.016** (0.003)	0.003 (0.006)	0.000 (0.003)	−0.008 (0.007)	0.005 (0.007)
[15,16) min	0.012+ (0.005)	0.022** (0.004)	0.001 (0.006)	−0.001 (0.003)	−0.004 (0.007)	0.002 (0.007)
[16,17) min	−0.001 (0.005)	−0.003 (0.004)	0.004 (0.006)	0.004 (0.004)	−0.003 (0.007)	0.008 (0.007)
[17,18) min	−0.002 (0.005)	−0.007 (0.004)	0.007 (0.007)	0.003 (0.004)	0.018+ (0.007)	0.007 (0.007)
[18,19) min	0.001 (0.006)	0.011* (0.004)	0.010 (0.007)	−0.000 (0.004)	0.005 (0.008)	0.006 (0.008)
≥25 min	−0.000 (0.002)	0.002 (0.001)	0.012** (0.002)	0.007** (0.001)	−0.000 (0.002)	0.000 (0.002)
Observations (years)	2,777,448 (2002–2005)		2,101,260 (2004–2007)		1,087,605 (2008–2010)	
Pr (1 min early)	0.18		0.21		0.19	
<i>p-values from hypothesis tests of equality of coefficients</i>						
Bin15 = Bin12	0.05	0.00	0.79	0.26	0.94	0.36
Bin15 = Bin18	0.13	0.03	0.34	0.79	0.40	0.70
Bin15 = Bin25	0.02	0.00	0.06	0.02	0.59	0.80

Notes: Standard errors are in parentheses and clustered at the arrival airport-day. Separate regressions for each carrier include arrival airport-day fixed effects and arrival hour and hub controls. Coefficients represent the change in the probability of a flight arriving one minute earlier than predicted, relative to flights with predicted delay less than 10 minutes.

** Significant at the 1 percent level.

* Significant at the 5 percent level.

+ Significant at the 10 percent level.

neither US Airways nor United engaged in threshold behavior either before or after they introduced their bonus programs.

It is interesting that we find a response to the bonus program in some cases but not all. The employee-level incentives created by this type of program are weak: the program is based on airline-wide performance, performance is based on a ranking and so is affected by other airlines' performance, each employee only works on a subset of the airline's flights, and each employee is only one of many employees whose effort affects the on-time performance of that flight. Given this, it is more surprising that the bonus programs influence behavior in the first two cases than that they do not in the last three cases. As mentioned above, we do not have detailed enough information about the programs and firms to determine conclusively why they affect behavior in some cases but not others. However, we suspect that differences in the structure of the programs (for example, which ranks they reward), differences in the perceived feasibility of achieving those ranks given the airlines' typical rank at the time of the introduction, differences in complementary organizational practices

adopted by the firm and any other changes that may be simultaneously occurring at the airline (such as a recent merger or contract renegotiation) would affect what impact, if any, the bonus programs have on employee incentives. This further highlights the importance of understanding complementarities between various organizational practices.

E. Implications for Reported Quality and Welfare

We have found evidence that some, but not all, firms in this industry engage in threshold behavior in response to the incentives created by the DOT's measure of on-time performance. This threshold behavior can take two forms: actual reductions in delays on flights that would otherwise arrive right around 15 minutes late or misreporting of the arrival times of flights that do arrive right around 15 or more minutes late. While not the focus of this paper, it is useful to briefly discuss how threshold behavior in this setting may affect consumer welfare. First, the behavior we uncover may distort the information that the disclosure program provides to consumers. This is certainly the case when the threshold behavior is the result of outright misreporting of arrival delays. However, it can also be the case when the behavior results from actual reductions in delays on flights close to the 15 minute threshold because the improvement in reported on-time performance that arises from a flight being moved under the threshold is achieved with almost no improvement in overall on-time performance. Given that in many months, airlines' positions in the ranking are separated by very small absolute differences in on-time performance, it is certainly possible that misreporting or manipulating the arrival times of even a small number of threshold flights could affect an airline's ranks. Furthermore, while we do not have evidence on how consumers use the DOT rankings, it seems possible that some and maybe even many consumers are not aware that the rankings are based on the fraction of flights less than 15 minutes late.

The second way that the threshold behavior may affect consumer welfare is by imposing longer delays on nonthreshold flights. This is of course not applicable in the case of misreporting but is a relevant consideration in the case of actual delay reductions if those delay reductions are achieved through a reallocation of resources from nonthreshold flights to threshold flights. We have explored empirically whether nonthreshold flights that arrive or depart around the time that a threshold flight lands have longer than expected delays, relative to nonthreshold flights that do not arrive or depart around the time that a threshold flight lands. Specifically, we have regressed the actual delay of nonthreshold flights on the number and fraction of threshold flights that landed around the same time and additional controls. We have also used as the dependent variable a dummy variable for a flight arriving later than predicted. Finally, we have examined pairs of flights that arrive at the exact same time where one member of that pair is a threshold flight. We have found no statistically significant relationship between threshold flights and the delays incurred on nonthreshold flights though, in many cases, the estimated coefficients are quite noisy. We suspect that there are several reasons why we do not find evidence of negative spillovers from threshold behavior. First, a portion of the threshold behavior we find results from misreporting and this will of course not affect the resources allocated to other

flights. Second, even when there are actual reductions in delays, this may result from utilization of slack resources or this may occur at precisely the times when resources are underutilized. Third, even if resources are diverted from alternative uses, any of a large number of nonthreshold flights could be affected and we have no way of knowing from where the resources were reallocated.¹⁹

F. Additional Results and Robustness Checks

We have carried out a number of robustness checks and extensions. We have replaced our carrier-arrival airport-day fixed effects with flight-quarter fixed effects and our results are robust to this modification. The results are also robust to two alternative ways of estimating the taxi-in time that is used to calculate a flight's predicted delay—using the median taxi-in time for a carrier at a given airport in a given month, as well as the median taxi-in time for a carrier at a given airport in a given month during the arrival time window. For the carriers that introduced bonus programs, we have explored whether there may be end-of-the-month effects. We have tested whether threshold behavior occurs only at the end of months in which the airline is close to achieving the necessary ranking for a bonus payment, but not at the end of months in which the carrier is far away from achieving that target. Similar types of effects have been found in the prior literature on employee bonus programs. Note that, in order for such effects to occur in our setting, employees would have to be informed not only about their own airline's overall on-time performance in the month so far, but also about the on-time performance of all other carriers. The DOT only releases this information with a two-month lag, so that the information would have to come from other sources. We find no evidence of end-of-the-month effects.

We have also investigated whether there is any evidence that airlines appear to systematically reduce airborne times in response to a flight's predicted delay at the time of departure, using an analogous regression procedure to the one presented above. We find no evidence that airborne times are systematically shorter for flights that—upon departure—are predicted to be about 15 minutes late. A likely explanation for this is that the delay prediction at the time of departure is quite noisy; thus the airline may not want to devote resources to specific flights based on this prediction.

Finally, we have tested the robustness of our definition for identifying manual planes by using an alternative definition which is based on rounding of flight delays throughout the distribution, not just at zero. Specifically, we compute the percentage of a plane's flights during a year that have a reported arrival delay that is either equal to 0 or is equal to a number that falls on the 5 minute intervals, excluding 15. This alternative definition has a strong overlap with the definition based zero delay and the results are robust to using this alternative definition.

¹⁹ For example, if a gate gets reassigned from a nonthreshold flight to a threshold flight, we cannot observe this in the data.

V. Discussion and Conclusion

We have explored whether airlines engage in threshold behavior in response to the incentives created by the DOT's approach to measuring on-time performance. We have found that airlines are heterogeneous in how they respond to these incentives and that some airlines, at different points in time, manipulate the arrival times of threshold flights. In particular, we have found evidence suggesting that airlines which self-report their on-time data misreport the arrival times of flights that arrive just above the critical 15 minute threshold. The threshold effects we find for these carriers are large—in some cases, more than doubling the probability of arriving one minute early—and completely disappear once these carriers switch to automatic reporting. These findings highlight that the form of reporting can affect the credibility of the data collected through a disclosure program and possibly undermine the quality of the information provided to consumers. In the case of programs which rely on self-reporting, the incentives for misreporting must be recognized, especially if misreporting is difficult to detect. Similarly, in the case of programs which rely on third-party reporting or inspections, the variability in how the third-party collects and reports the data may undermine the quality of the information (see, for example, Becker, Jin, and Leslie 2012).

We have also found that several—though not all—of the airlines that introduce employee bonus programs based on the airline's performance in the DOT ranking engage in threshold behavior after the introduction of the program. This finding is perhaps surprising given that incentives created by these firm-wide schemes are weak. Yet, this finding highlights the fact that how a firm responds to a disclosure program will depend not only on the incentives created by the program but also on how those incentives interact with organizational variables to influence individual behavior inside the firm. Disclosure programs rate firms; yet quality is often improved or manipulated by employees who may or may not care about the firm's performance in the disclosure programs. Firms may differ in the extent to which they translate the external incentives arising from a disclosure program into internal incentives for the individuals within the firm who are in a position to affect product quality. This may be an important reason why firms in the same industry subject to the same program respond heterogeneously.

Overall, we believe this paper begins to bring together the economics of disclosure programs with the economics of organization. While we have been able to measure firm-specific responses to a common disclosure program and identify several firm characteristics that are correlated with those responses, we still do not have detailed enough information about the firms and their practices to precisely determine why they responded the way they did. Going forward, we think there is value in combining data on disclosure programs and market outcomes with as detailed as possible data on internal firm characteristics to better shed light on whether and how the incentives created by a disclosure program influence individual and firm behavior.

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